

Conclusion (Discussion) ~ Active Inference for the Social Sciences 2023

Source: [YouTube](#) **Playlist:** Constructing cultural landscapes: Active Inference for the Social Sciences (2023 course) **Duration:** 1:26:58 | **Uploaded:** Unknown

Transcript method: auto_caption

hello welcome everybody it's November 1st 2023 and we are in the closing discussion section for this first instance of active inference for the social sciences it's been a journey I'm sure we'll have many fun things to talk about and explore today so aii please take it from here and maybe give an overview of what we're going to do and talk about today okay so this is the uh conclusion session of our course on active inference and the for the social science we have talked extensively about active inference maybe less extensively about for the social science parts so the goal of the session would be basic to assess uh what we talked about and how it is relevant or not to the practice of social sciences um basically we will uh this is a discussion session so there is no clear plan but I'd like to address uh how formally active inference accounts for agency and normativity and identity as a kind of integrated phenomenon how it uh can compare or add to or ground or clash with I don't know a sociological anthropological model of agency and of social evolution and of etc etc all the topics that we addressed which are numerous and discuss about the plausible future for active inference what it can be used to uh maybe what it will be used to certainly how it may change and how it's relevant to the discussion so um yes that is the um the uh Crux of it uh we have brought in Alexi who is a philosopher interested in social ontology specifically the study of social kinds which is from Chao's research and I do not know who Colen is but he's with us hi Colleen uh so how should we proceed Danielle do you like what do we do if you'd like maybe pose a specific question or prompt which could be preceded by a statement and then we can all hear everyone's voice on that okay so a good question would be uh active inference uh describes agency and normativity and identity as a kind of um integrated phenomenon I think we should go over how precisely it is uh it works um so yes let's do that I will uh find some do you say illustration that will make it easier but let's hear people's thoughts on that please can you pose your question just a bit more clearly how how are agency normativity and identity addressed separately and together in

active inference now go for it um so I think we like to look at things in terms of inter relations in active inference nothing is defined by itself it's defined in relation to other things and this definition in relation to other things also defines the types of constraints that will be pushed on the thing which is separate from other things um once you have this you necessitate these inter relationships to be formative ontological and um fundamentally contextual so nothing no relationship is um perfectly reproducible across a different context it will take on a hermeneutics a semantics given the exact context of the spatial temporality of the entities um those are some first thoughts does anybody have like big objections to what I I just [Music] said okay so let's start oh yeah please continue sorry let's start with the concept of agency for instance um agency is just action selection right so uh agents are are modeled they have a generative model uh and they use it to predict like sensory inputs based on possible actions um they they try to select those to minimize the difference between uh predicted and sensory inputs uh so they're just trying to minimize free energy and this minimization is fundamentally due to the relationship of constraints in their environment so let's imagine that um I am an individual who um oh sorry I saw a comment and I didn't um so yeah so you you uh you have an individual that tries to exert control over the environment now if you scale this in nested structures uh you can try to think of power structures and how agency within those power structures is modulated by the different kinds of constraints that are both embedded within the environment itself within the materiality of the environment and also within the the normative aspects of the the nested structures Which pull and push an individual um within certain paths through some kinds of Deon CU so that's how you get to uh the the the Notions of encoding agents beliefs and preferences about the world within the generative model through a form of normativity which is just a higher Precision um over a specific type of um outcome or mapping so uh your your prior beliefs here come from different layers in the hierarchy and these act as what we call culture or normativity um I could stop here if you want to say something a so I uh agree with what you said at a high level conceptual scale but I think it's important to uh be clear about what formally the statement because what we have formally is much more limited than what we agree the interpretation could be so can you see the um the screen I'm sharing can you see the mouse doing AIDS yeah okay so at core uh active inference is a framework uh that provides geometry for gical systems that is that is not a very um straightforward to put it but what we have is a system with a space of states and uh we have a flow of the system a flow is a function that basically tells you where a specific point goes uh with time and what active inference says is that given um yes given specific patterns of demal

coupling and of statistical dependencies we have agent like Behavior so the simplest kind of the basic requirement of active inference of sorry the principle which is a core motivation for active inference is that you have anical system so something with States and flow and some noise because we need noise to get information and it has a boundary in the middle of it that um statistically separates the X and ϕ in states by statistically separate I mean that every information uh about what's going on on there uh that is in there it disappears when condition on there so information I have about this that I could read here I can also read there and that's sllly different fromal coupling but this is formalism let's not get too much into this and then uh you have more complex behavior when you can partition sensor in active States and your particle can do things in this specific context we can uh write basically we can write a geometry for the fluctuation of the system and this geometry it entails belief of external States so this internal States it knows thing in a sense about what's going on out there through its coupling uh through sensory and active States and the flow it hases internally the weights internal States move they entail um um a displacement of their belief and this displacement happens to be near optimal given base constraints blah blah blah and the real interesting stuff happens in the case of strange particles which are particles where uh as you can see there is no direct effect on of uh from actions states to internal States uh the only action you have is mediated by sensory and external States in this case uh the active States themselves so the policy of an agent it becomes uh the the subject of beliefs of the agent and if it becomes the subject of beliefs you can have statistical inference this is the Crux of the framework and if you can have cical inference you can have like quite complex multi-level Cal inference over re actions and those are basically what we call decision taking or what we call planning so um what we have here is a formal architecture that can derive the kind of complicated um agency uh semantic not only agency but uh agency that is embedded in a specific conceptual framework that is relative to the specific structure of an agent and their environment and the coupling in between all that is Trace back basically to Dynamics so we have a proof of existence that in a world that everything is aramic we can have agency we can have complex decision making under the form of inference over one's own actions and then you have complications to apply it directly of course because um the world is likely not in its bar formal system it does not have a list of states that are um okay uh there is not a lad of states that are possible and that's it and there is one coupling that takes L stes to themselves and nothing can change ever this is not likely to be the case uh but we have a minimal model of agency that is purely physicalist that is purely based on Cal relationship in the most minimal way we can

possibly represent it and this is quite huge and this tells us this this gives us a very very strong prior of what we can what is the stuff that agency is made of and the stuff that agency is made of at least higher agency that we like think about here in the social context is belief of oneself and so to control actions is to control belief of oneself and this is this um this Duality this like dual aspect of uh statistical organization as a complex cognitive agent that is is in my opinion at the moment given the capabilities of modeling informalism we have the core uh contribution of active inference to social sciences written large in Psychology NEOS science like science of social stuff because it gives you a very strong prior over uh what agency is made of and the kind of stuff that ex that can exist out there in the social world and the kind of dynamics that uh bring about it stuff so but there is a quite uh big bridge to be made between the pure demal system formalism of AC INF frence and the kind of evolution that happens out there and I I do think there is um high level of precaution to be taken when we talk about this kind of stuff and yes does anyone want to react or agree or disagree violently because I just attacked the uh common creeds not sure I perceived an attack perhaps um from what I understood of what you said basically um you have information geometry it formalizes a belief space in a sort of geometric manifold uh the points represent possible belief systems that an agent can have and uh the distance between the points in this space can be interpreted as a measure of how different two belief systems are and this is necessarily um tugged by the the priors given by the environment which also push uh uh the agent to act in a certain way given the type of thing that it is in order to maintain itself given the the actions that it knows it has available um and this becomes normative because of the um the strength of uh of of the attractors within the belief space which become the geometric landscape basically um so we can sort of uh model the the the attraction through the curvature of the of the space around the attractor and so a high curvature means a strong normative force uh which pulls the agent to adopt a certain I like to say social scripts because that's what I've worked on but you know any sort of policy and or phenomenology Norm um and so in this context the agent agency is kind of complicated because it's both um the ability to sort of navigate the landscape to move closer or further away from the attractors uh and so the degree to which an agent has agency is a degree to which it doesn't go necessarily um directly towards the attractor but has some degree of reflexivity over its own um action space in order to navigate that uh the the different kinds of um uh tractors that may be available to it uh so we could see that the momentum would be like affected by uh personal experiences education in Tendencies uh or even you know the tendency to to resist or yield um to uh to social pressures I could go on but I know that um a

you have your hand up so why don't go uh okay uh I guess a could answer before before I talk a bit so I was just trying to have my hand raised before um yeah before talking so if ever wants to answers you and afterwards I can talk uh I'm okay I don't have anything specific to answer okay okay uh so you you talked about the dual aspect of statistical organization of complex agent uh and you said that was a core contribution of active influence to sciences um can you dig a bit on that um and yeah Ju Just articulate it a bit more so we could grasp it better and then I could uh ask questions if so do I still share my screen uh yes so basically what we uh what we have in here is a formalism that um that builds from very basic Dynamics in like just space and there is stuff that to paace zero level physical presentation of literally anything can go there and uh from there we build a model of first uh self organization not quite that agentive self organization let us say something that can build a specific stable ecology and maintain itself in this ecology that is what we have here and then uh given the lack of coupling between active and to States we can build uh sensibly more complex stuff that can Implement uh higher order beliefs about themselves and what they do and uh like do decision making that is condition and information they have about themselves and planification under the form of uh beliefs over their own action and as such this is like this is just math this says nothing about the world in and of itself until we claim it corresponds to things in the world and we can claim that uh this is a basic conceptual model that Maps well on culturation basically social organization engagement with the social cultural world because then we can account of uh the human ability for very very very um white open-ended learning as something that comes from there as something that allows us to infer things about ourselves and then develop complex policies and knock them blah blah blah and because to my knowledge there is no competing model that that comes for anything near that like the the only competition like the competition I should not talk of competition but the um the other models of um cognitive organization curation that exist that are trying to ground themselves in physics like Nal system they are just talking about Dynamics they are specifically avoiding talking about anything like content of or semantics so the facto this piece of math as a monopoly on building semantics from the ground and what it tells you is that agency and identity and relativity it's all the same phenomenon it's conditioned by a self-belief basically that is constructed in this specific way and we can um how to say because this is the only way that we can account for higher order agency then we have we at least I would argue we have we should have a strong priors for uh theories of agency either mechanical or uh conceptual that Vibe well with that picture so this is the claim I'm making I guess I want to too much detail so I hope this is enough okay um okay

so I have some maybe not questions but some points to make now um thank you um so from what I see here uh we could put the active influence Paradigm into the individualistic uh methodological individualism um framework in sociology so the idea is that you try to model an individual and then uh when you have the good way to model the individual you can generalize to what everyone is doing so for instance um if I'm working on something uh in certain way then I could generalize about that and say uh okay sociologically uh I have this constraint to work on this uh stuff and then not like everyone is doing that but it could go like that but what is interesting here is that it's not only about one individual but also that uh for instance let's let's take the external State you could say that external States here or other agents and also that um okay so you have the active state in the internal state so we could say that your internal states are totally influenced by another agent so we could do something like that with the two individuals I guess so one is one of the agent here and the other Drive the internal states of the model because they are really influential on uh on yourself but so there are some limitations on about this kind of way of thinking because in general um in sociology or in sociological Sciences what you would like to say is that there are some agents at a higher level than only individuals such as you and I so the trick could be to say that some states for instance or some firms or agents uh for and by themselves so you have agent which are not individual persons uh about you talko social scripts I don't understand what you tried to say a um I was just trying to react to your your slide um because I don't see how it connects with my own work here yet oh I'm talking to M okay nice um so yeah if if Mar you have something to say about what I just said about how we could have let's say two active agents uh trying to talk together uh yeah so um we we have such simulations that already exist um over various kinds of scales we have um two agents coordinating um that are animals like birds singing and overtime coordinating over the song they're going to sing uh we have uh several models of leader follower where um you have one agent two agents starting from relatively the same point but one agent developing more confidence and therefore developing a dynamic of leader follower with the other agent this is work that's possibly um going to be followed uh by Francisco Balan who is working on um models of education and how maybe the teacher and student and dynamic may be cast under these terms um I've done work with Axel constant and maxel ramstead in terms of um active inference in social scripts that's that's what AA was mentioning and this notion that um through the cues of the environments your own model is shaped both by the environment which includes other people and the way in which you self evidence within this environment so for instance um there's two ways we can understand your

question first like we can say that um U minimizing free energy um is basically an agent that has a form of autonomy and it's constantly updating its model to align with the environment so by virtue of doing this it's not fundamentally individual but um that each agent has its own model of the world each agent reflects a perspective a very unique moment in SpaceTime that will um that will incur some errors relative to others so it's not that they necessarily interpreted things wrong although they might have but um because their perspective is slightly off from somebody else there's necessarily a degree of incommunicability in this sense um so here self-evidencing really entails that uh um identity you create a sort of identity here through the fact that you uh you may have different preferred States from somebody else based on your values belief your social conditioning but again these beliefs and values and social conditioning have been shaped and and um again we're talking about these 12 of attractors by other people um and so we try more and more to challenge this notion of individualism by emphasizing the relational aspect of identity um which is which means that identity is not pre-established but continuously formed uh through the interaction with other agents uh and again if we go back to information geometry the the the geometry of the information space influences how agent perceive their paths towards minimizing free energy and affect their choices which again give you uh the possibility to understand um how agents might be very strongly pushed by other agents let's imagine an agent that has more power than you and is uh trying to influence you to do something someone that has very high Charisma or someone who has the power to say destroy your life you are more likely to shape your belief beliefs to align with and therefore we can understand individualism under a very very different light here um AAL did you want to add something Colleen is waiting to speak since a moment so let's see Colleen hi I can you hear okay I just I just wanted to pick on a couple of the things that Mah was talking about um and just now and in her in her talk it seems that uh the you you seem to be saying that that for social conditioning and the way in which we are socialized uh the the importance of social scripts appears to be greater than the influence of individual upon uh individual and that's interesting because it's not something that we would tend to think as individuals that we are we are integrating the scripts in fact the the whole idea of seeing a script as something which is social uh was something that I found that probably most people would find uh a rather strange idea um off the off the top of our heads so so yeah that uh is that a fair is that a fair assessment of what you're trying to say and the other thing I wanted to ask about was uh a few times uh in in talks recently and here today um I've heard people talking about identity and the active inference view of uh of identity I'm not sure that I really understand what

identity means within an active inference framework as distinct from just our internal model of who we are uh is there an other is there an other idea that we are we are uh we are we're thinking of here and more particularly the word identity has a lot of baggage and usage in current contemporary Society um are we using it in a different way uh and if so how is it different and potentially how does that give us insight and the ability to engage with the ongoing discussions in society in a potentially fruitful way so I I think I lost you for the last part of your question which seemed to be about the difference between our conception of identity and the the conception generally used if that's uh that's correct you're muted calling sorry yeah I was I was referring to the concept of identity as used in society um right so I mean I'm not going to speak to every use of the term identity because it's a very very useful term that's used in law that's used in physics it's used in sociology and psychology so it's used a lot um all I can do is explain roughly how we consider identity formation and then if you have a specific definition in mind for a specific field we can maybe tackle that one does that sound um okay yeah um to be clear I was thinking specifically about identity as determined by group membership in sociology um okay um okay so I'm going to tackle social scripts first yeah um um I'm going to tackle social scripts first and then I'm going to come back to identity because I think it fits I've worked on that as well I know that Daniel and a have some work that relates to this as well but so your first question was um on the importance of social scripts and what it means for cognition if I understood uh correctly so from an internalist perspective um social scripts are cognitive structures they guide behaviors they're they're acting like um brackets we use the word constraints a lot it's it's not a perfect um it's not a perfect guide because no semantic marker is a perfect guide everything is up for some degree of interpretation it's just that there are some interpretations they're so strongly linked to some degree of embodiment that they sort of anchor the rest of our semantics so for instance I can't stop breathing if I wanted to CH I I have I will stop but eventually my body will restart on its own um and so there's there's such kinds of natural laws here that um create these these um bottom layer guidelines that can then be sort of derived to infer other things and social scripts are just a representational um social extensions of these constraints given a historical context so um they model uh and did M drop or did ey drop I guess it's given to you like somebody will actually tell you what to do and show it to you um and sometimes they won't be sometimes they are just things You observe sometimes there's things you infer because you made a connection between a week script and another week script so if you see something that resembles a chair in a room you infer that you have the affordance to sit on it so that's a form

of script then you see other people doing the same in the script sort of cements and so now um You you associate this to another part of a sequence and this becomes something that's um that allows you to temporize um and and basically leverage your B Matrix in order to exist in the world in a way that doesn't cause friction with other people this also gives you action triggers there are things you see in the world that now tell you this is the thing you should do now so there's this sort of um recursive Loop that's very inactive that pulls you in a Direction so the the attractor isn't just a state it's a path you're going to go down a path um so for now is that does that answer the first question yes thanks that's helpful okay uh ahead thank you uh to get back on the like you talked about the active Loop and I think that's a critical thing to underline which is that the engagement with the world is a cyclic process a circular process it construct its own conditions uh and the so there is no measure of power of structure versus Power of Agents because the structure is embedded in the agents and the agents are embedded in the structure so I I'd like to share my screen again uh to give some detail over what this entails do I can I yes good window you should see a PDF with a tiny man a personal Mon and Loops is it okay it's okay yes I see yes so this is a presentation I did recently about the uh unification Church which is uh like when you want to be uh published you say new religious movement but you can say cult it works uh which originated I think in Japan and is now spread between Korea Japan it says mostly we don't have numbers but it seems uh to be the places where it's most implanted and it was very very very um remarked because of the personal cult around the Reverend Moon the founder and the Very overt exactive practices of the cult like asking you for the equivalent of 10 years of salary at the time to uh and to take your ancestors from hell and put them in Paradise by giving money to R Moon specifically and they they were the topic of a very um I say aggressive and excessive anticult movement in the 70s 80s and now they are becoming a bit noticed Again by Vice uh not only Vice but notably Vice because you have very strange strange things like Mass Maga weddings with people waving rifle and that is a very strange organization and at the time you had the very um moral Panic with say know around the cult uh you had you had basically a model of influence that was that evil people would brainwash you and then you were commanded by the cult and our Soviet secret agents from afar and they could like say to you horse and is triggered uh homos cidal Mania in which you blah blah blah so things that are unnatural istic I will say and uh notably in reaction to the moral Panic you had a movement by solist of religion that would try to understand how exactly cult worked and the most famous sociologist to do that was aen Barker if I remember well she uh like she was she warned the moies she was coming so you

can question her methodology uh but she seems to have done the most thorough and uh well received analysis of them so it's quite important thing and she questions that there is such a thing as brainwashing there is a construction of a specific social context that you have to comply to if you want to come but they can force you they they do not have the capability to uh like torture you and make you into a RC radio Community thing that's that that does not work and so I forgot the name of the so David Frank Taylor uh explained uh the influence of the church in terms of choreography of to participation so basically you have very huge very um emotionally investing ritual that have you just have to take part in to be a member of the community and if you are a member of the community you will be helped you will be loved you will be supported you will be uh constantly encouraged in a slightly coercive way to uh invest more of your time and money and whatever in the organization and if you don't then all this sport just leaves because you know you're a traitor you you you have you have uh done uh treason of the true parents who will Who was kindly out of his own art trying to uh save you like Jesus did but better because Jesus failed by not having children that's that is very dramatic thing to say that is what they said and because of the way the organization is set up there is you you cannot possibly question that your questioning is not accepted there is no private time in which you can privately question yourself you are always engage with people who are either uh formally or informally depending on your seniority watching you and so you have basically a context that makes it very very very easy to align with the uh cult values and it just happens that those values they they entail self-sacrifice for the organization but also for new members and uh total difference to the cult and its leadership and anything they tell you to do so you you know it's not nothing but it's also not exactly free consent to a specific context and from a purely uh I don't know what to say mechanical perspective they can they can make you agree when you are there but they cannot make you go there so a question is how precisely they can you know elicit a compliance away from the direct context of the choreography of participation and there is a like simple answer in my which is that U we have we have a coherent sense of identity we can't we can hardly think ourselves as a good money one minute and then step out of the meeting and not think of ourselves as a good money that doesn't make sense either we are or we are not at least most people tend to have one integrated sense of themselves as someone who is embedded in a specific social context and specific norms and behaves in such such way and so the uh the coherence of self-identity it provides a key coupling between your public and your private beliefs what you are expected to do in Social context and what you will effectively do and perceive the world as privately and this I would

say uh in the example of theonism but in other rhetorical examples like uh the emergence of nationalism it is a very straightforward confirmation that indeed coherence of self-identity like the coherence between what you do and what you think of yourself as doing is a very important process in the uh in the structure of societies generally speaking at least contemporary societies maybe it was not the same before AI so I do not think like the context in which it's moving of course it changes but the mechanism I don't think it's new and I'd like to get back on Alexis's point that um sorry A's Point that's uh the approach we have is um individualistic I I do not think it is the case because if we are correct uh there is a landscape of social cultural affordances constraints uh things world out there and we engage with it and there are mechanism that makes this landscape not necessarily identical but very coherent between individuals and those mechanism are language and the material symbolism developing ritual which is also symbolism also meaningful but maybe less than language I don't know uh I just mean that the mechanism is the same and uh the um interculturally robust practice of Role taking and uh Shared intentionalities are trying to do stuff together um given a specific uh model of the attribution of tasks within the shared uh goal and so these are human EnV variants and all would be pretty good candidate to force the integration of a common model of material and cultural Niche and so I think that there is a pretty coherent sociomaterial niche I think that uh we can we all engage with it and we all modify it and we maybe do not perceive it as the same but there is still a strong enough constraints uh on the coherence of this landscape between individuals for it to make sense as a specific entity and then the proper respiratory scale of what we do is not the individual Minds it is landscape that we happen to experience because of our minds so by the um PL uh Plateau Academia of Stanford on holism basically holism is a theory where you agree that there are uh explanation that are not mediated by IND States and I say there are explanation that are not invented by IND States and not only that but I can point organization that control people while being absolutely not worried about individual states only creating a common text and then allowing them to engage with it and that would be compelling in my opinion to say that the model we have is Maybe not holist maybe a secret third thing that is multiscale but certainly not individualist at least in the sense that you are mobilizing so thank you uh I talked more than I wanted so you were raising your hands I am but I think Lorena wants to go I will go after Lorena hey guys uh yeah I think that goes like directly into the idea that coming a little back coming back a little bit to the point that identity they were talking before and if you think about the model that we have uh we have external the external States and different agents they are always actually very local in

situated in the same um in the same local constrained local environment right so your external agent is going to be part uh other agents are going to be part of your external States and you are part of theirs right so the idea of like laying down a path and creating a specific type of dynamic that react to the same type of input that you're receiving and driving these Dynamics will generate this sort of embeddedness right because you do create you do have the same um or it's not necessarily the same but they're very very close to each other they're very approximate right so the idea of a dynamic here if you think about uh describing that in terms of um place and like in terms of Dynamics you do have very different like um Lauren but butterflies for instance each one goes through the same type of Route however they're not identical but they're very similar they are doing something very similar and that similarity sort of constraints again uh the construction of that environment so if you think about symbols and roles and uh types of environment that you find they're going to be as concrete as something else that you find in your environment constraining and pushing you towards the same direction or at least a very similar uh you can describe dynamically in a very similar type of geometry if you if you like or a very similar or this attractor will drive the very similar type of uh path we describe this similar type of path so which goes exactly to the uh a can you go back to that um other um yeah the the other figure that you have the one with the enviro M as well made by yeah that one so that exactly that creates like a dynamic that they can be then for forgotten like in a way that will be described through a a specific Mar of blank that will develop agents uh together we push them constraining they're going to be constrained together and they're going to have a constructive kind of um role in this embedded uh In This Very embedded normativity but yeah you go a thanks um yes so I think there's a nice way to tie all this as well to the internal nature that we seem to be um dispelling but we're not we like we have to talk about specifically in what ways internality is cast here and um I like to think of that the the effect of self-esteem here self-esteem to me is the degree of Fitness relative to your perception of how well you are managing to exist Within These constraints and interpret them in a way that gets other people to validate and and minimize the friction that you're getting from other people so uh it it aligns with the concept of sociometer theory it Taps into this notion of cult um so if you consider that there's effective states that enable an individual to track and respond to discrepancies between self-perception and social expectation through these effective states such as um anxiety or positive affect um we can see that there's a dynamic nature and self-organization of self-esteem which is continually shaped and reshaped by experiences and interactions um and so uh under active inference we can formalize this by

saying that a model uh that is more precise uh effectively captures the higher order structure of the moment to moment variation of more stable traits such as uh what we consider to be self um and so basically um we can see that identity and group membership here are fundamentally related to the effective experience of um Being Fit for the group membership and so once you're there you can grow in Social cohesion and this will lead you to potentially try to maintain the group it will try to it will lead you to try to justify the beliefs of the group such that uh you have a form of confirmation bias but also you maintain the epistemic reality of this group The symbolic reality of this group which would be more costly to destroy than to actually uh support it's also integrated into your self-esteem therefore destroying the group would mean admitting that you are potentially not fit for the environment that you are uh potentially some degree of wrong um and so this this um this leads us to the possibility that if someone has low self-esteem what do they do they can either stay in this group and accept that they are unfit for the group uh and they would try to remove themselves to some extent from the group to avoid all that friction or they can try to join a different group one which will support uh your beliefs about the world if your beliefs are too far from where the the group you're supposed to belong to currently is and this is where we get to these subclusters forming these little uh subgroups uh that potentially can even become um sometimes violent and sometimes really not sometimes it's just you know some tumbler group but um it has power because it gives the individual this feeling this veilance um of now being fit to a group that supports their model so I wanted to give AA space to talk yes thank you um so I would like to go back a bit uh on what I said about uh individualism so yeah I agree that uh I was wrong about associating active insurance with individualism because what is interesting here is that there is an individuality through active inference which emerges from uh the collective and you can justify or describe uh individual actions or else through the L uh through the landscape as you did just now and then go the other way around so we here we seem to grasp the interaction between the individual in the group and vice versa uh which is what I found interesting here um uh and okay moreover there's the what the individual adds to a group uh I see that here it seems to be captured as we take the individual as a um how to say that uh a unique perspective this is what you said before ma uh that uh each person is a perspective uh a unique one and yeah uh it's not all what I wanted to say also uh yeah Lorena you talked about the the beliefs that exist and I think this uh this is one interesting thing also in active inference it's that we take belief as a material thing that uh influences us and that we can also influence or create in a way um even yeah independently of

if it's right wrong or whatever we we judge it is like uh what we talked about with the moon sect uh and the the unification Church yeah so there are people with crowns and we say that they are the big boss and we have to deal with it or to go away from the landscape and from the landscape and yeah it works like that in the social world indepently of if the person is really the like the the the chief of the Cults or or not because they said the words and people were just following them so in the end it happened to create a landscape in which the in which one person is the uh the the person who has power so yeah I don't know what to say more about all of this I guess I agree with with what you said just before so yeah that's it so maybe we should move to like challenges and opportunities of active inference uh so uh two points I think like I just put them out there you can react um I think that one active influence may give us computational models or even like thorough gred conceptual models of social organization specifically the like semantic Dimension to social life and I also think that in its current formulation within the M System Theory it is inapt to represent par Evolution because M system entail the formulation of set of states that do not change through time and AAL flow that does not either change FL time so there is there are opportunities to apply the model to the framework to uh grounded meaningful semantically complete models of soci evolution but this model if that come for evolution they will not really Vibe well with the math that we invoke to say our computational model are meaningful and that's like rise and AR R of questions I guess so yeah does anyone want to react to these two statements could I go sorry could I could I make a comment um just more more specifically I'm I'm I'm interested in the in the ways in which uh the ideas of active influence impact on the way we think about society and the way Society Works uh and I I can see from from my reading and from listening to the disc discussion that there's a very rich du lexicon in vocabulary a set of concepts for the thinking within active inference uh thought and as we just as we just commented it's quite heavily mathematical so the barrier to entry uh for for for the full active inference uh um worldview is is over time these ideas will will re will permeate out into the mainstream and they will change the way people think and talk about Society I I'm interested in the ways people who've studied a bit more than than I have see potentially how this will change the way Society works or the way that uh the way that we we uh we have disc we have discussions or Envision Society more widely than just an academic discussion obviously um maybe the question is a bit too big I mean I think I think everyone is just waiting for the other to talk first talked first so um so I I I think I want to tackle a little bit what you said about um the uh the open-ended uh nature uh the open-ended Evolution uh of of um of State spaces

basically um I think under the path formulation um this open-endedness is actually just a perceived open-endedness it's just um a function of uh the time scale being considered and the motion along that time scale um I think uh I think we can go a little bit more into depth about this um I think it's a very large topic like effectively um if we go into Quantum formulations there's um some degree of irreducibility and anything above that is historically contingent so obviously some things will some Mark of blankets will dissolve and reform and dissolve and reform and through this dissolution and Reformation uh you get a new potential for mapping of observation to States so there's there's always that possibility um but I think some challenges um for active inference that are coming uh is for instance uh the the complexity of social behavior um you know we we um we can extend active inference to capture the nuances of interactions uh but effectively active inference is about you know abduction so it's the best guess for now um and it requires for us to put into the models uh what we think are useful variables what we think are the right variables um and it's possible that we're always going to fall just a little bit short like we're always we can either throw a bunch of data at a problem and be like okay well just just remove the noise only pick up the signal and figure out what's happening but that's no better than than big data deep learning so um we have to figure out a way to move forward and understand the the variables of interest and do this across scales so identify the priors that take over at a given moment but they won't take over even if they're higher in the hierarchy at another moment because in this hierarchy you've actually switched States like you've moved uh and this is only predictable uh once you've understood the effect of those PR PR um you can just consider that you have the priors to begin with um so yeah as far as uh uh integration with existing theories I think we have a nice opportunity ahead of us it's what it's basically what you said um Colin I think we're going to be able to more and more through um the alignment of certain kinds of language and the formalism um under that language we're going to be able to pull more and more theories and find what's at the core of the connection between these theories active inference is supposed to be scale-free so it describes or it is possible to help us describe the phenomena basically from sociology and literature to biology so if we understand this we understand that there's a possible touch point a convergence of all these fields that are tapping into to the same ideas or into some connecting points that allow us to understand how we maybe do away with these um different fields remove the silos and come back together with something that feels more integrative um but then again we're back to the computational limits obviously like if I want to compute um the prediction for an entire country what is the granularity level at which I I should compute

the the right variable and if we understand that uh individuals are formed by the group Etc do I just do a mean field over the entire group in which case am I losing out on some of the variables that may actually impact the outcomes um I think these are the the key questions that remain to be answered to some extent so I agree with the what you said now um I I agree with that we can't like represent things and not somehow not cut anything relevant to the thing when we model or we represent the point of it is to have a more compressed way to engage with the thing with our knowledge of the thing than the actual thing so we will just cut information that is the point of modeling or representing and uh maybe in a very simple system such like um just say perfect gas model in physics there is information we can cut and lose nothing we can uh cut information about which particle is where because the different particles of the same atom as at this scale the same properties so we don't really care where atom what what Atom does what but in the case of human societies we have like patterns of Engagement and the like specific information that is accessed by spe agent is important important and whatever we choose to cut for a model it will be relevant at some point at least it will probably be if like the information we cut uh is not specific to a bunch of agent that will die of car or something uh if any kind of information any type of information we may want to abstract out will be relevant somewhere at some point so we yeah we we are dealing with complex systems we are not going to model everything but I it is it is it is a huge deal that active inference is capable to fill in the semantic content from Dynamics either from like the Dynamics of a component system say neural Dynamics and biological Dynamics or dynamics of a social context with um the modeling of act influence agent as a like things that navigate the world and make inference and stuff and to my knowledge we do not have any other theory that makes the bridge between like just Dynamics and semantics and we have other neurological like cognitive Neuroscience people who talk about Dynamics a lot and how we can bring back uh the activity of cognitive systems to Dynamics but usually it is um it is in a way that insists that we we should just look at the Dynamics we don't really need to worry about the semantics or if we need to worry about the semantics they somehow like reduce to the Dynamics the Dynamics is what exists and is important and what we should worry about and so another framework that is relevant to like just study evolution of norms or presentation or mental content in a context I don't I cannot think of one one that is Thoroughly grounded in a at least that that attempts to thoroughly ground itself in mathematics mathematical physics I yeah there is the isent another one and so that is a huge deal but there is a very important problem which is what you what I said and you said and we which is that the paress in the path

integral formulation in only ofation in the FP it is illusory it is about like it uh so maybe to give a bit of context uh the path integral formulation of the FP is just when you do the FP so the of math we make huge deal about and we uh apply it not to uh like the Dynamics of a particle per se like how it affects a particle at Point T but how it affects pathes of a particle for mathematic reasons uh reframing a problem in path integral formulation is a huge deal it is what what makes the difference between a Newtonian you push something and it moves and full-fledge quantum go theories that uh underwrite all of contary physics uh so yeah path integral formulation is a huge deal but to write it you still need to write an AIC space like to make the list of all thing that are possible and like if you if you think that there this is a correct model of the world and a complete model of the world and this is everything there is you have to agree either that caes do not exist or that there is a place in the Primo web function that existed before Universe did before time did there is a place where con lies and I think Co exist and I do not think Co are basic properties of quantum mechanics So Co must have happened along the way somehow and to represents that mean to represents upness and so we need upness in math and the FP is not one ended and it like structurally it's very hard to account forend evolution and at a more praic level when we look at engagement with the social dynamics with the social world we look at piness because there is always a integration of norms there is always a loss of information but also that is constructive of new Norms there is no like writing happen at some point and then we did things differently it changed the structure of socities it enabled Administration to exist as a non-trivial uh C power Society that's actually mattered in its organization and yeah so new things happen and if we want to look at social engagement it is critical to look at how new things happen and currently the FP doesn't do that and active INF frence I don't think it does that maybe maybe it can at a computational level without uh Grand from all pretention maybe but I did not see it and that's a big challenge that we have ahead of us and by we I mean everyone not just AC INF frence people everyone should de a that at some point Daniel you were raing your hand I I I'll add just a few notes to Colin's question certainly it's something that we all ask ourself and many viewers and participants in the ecosystem ask like so what what then what does it really mean for active inference to be applied to or by or in the social science so I think that's a big open question certainly one that's not only addressable by one perspective or even with words alone but I think there's a few key pieces especially that differentiate it from just the broader or adjacent project of quantitative social science or formal social science the the question we should seek after is what does active inference bring in and again probably many ways to say

it but we know for an active inference agent as as as a scale-free statement free energy can be bounded by changing the world or changing the mind so having continuity of social systems with all other systems spatial temporal arbitrary material and so on gives a continuity in an interoperability that without a scale-free first principles approach is essentially not even on the radar so that's one massive piece and then I think where that begins to bear or modify fruit is essentially the scientific and professional development of cognitive engineering understanding that the cognitive constraints and scaffolds scripts all of these tasks implicitly and then more recently explicitly and most recently quantitatively and formally are examples of cognitive engineering so whether it's a stop sign and what that comes to mean in different social locations or or language as a complex phenomena these are examples of cognitive Engineering in the same way that building a stack of rocks is both an example of material engineering and cognitive engineering through semiotics and now when the scientific process the professional process enters into this reflexive and relational posture through active inference that is kind of a saddle point into a different Dynamic I believe Lena Lorena unmute and then go for it yeah I just make uh one comment what you said like keep in mind that active inference can understand beliefs and signs and this kind of of um semiotics as material and concrete because constraints they constraint the evolution of the system like how the system complexifies right and the constraints are materi has this material uh effect on the Dynamics right you cannot really um believe that of course belief is materialized as a table of course not but for the system in terms of constraining and understanding how things is SC and how they complexify they have that um kind of role and that helps to understand a little bit what Val was bringing about how you can bring new uh values and parameters for the Dynamics that are being described by the mathematics of external and internal free energy principle uh updating right and that is a way of tackling this um novelty for this kind of um this kind of thinking over time but we don't have a proper account on how to how to predict what is going to constrain the the system last right yeah Ma go for it and then we'll all kind of give a closing round slash thoughts okay well there was a um a question in the chat uh at one point so I just wanted to potentially address the question if that's uh if that's okay um so the question was so would the generative process of modeling the social system enclose multiple agents on one uh Mark of blanket or I assume inside or under one mark of blanket so we like to think of active inference as nested in hierarchical that's one thing that Maxwell ramstead definitely pioneered uh where each agent has its own uh Mark of blankets but there is a higher level Mark of blanket that encloses the the social system in play so individual agents can interact with each other and the

environment but the social system as a whole also interacts with the broader environment and this is where semantics and semiotics come into play uh and the narratives around certain groups start shaping how they can interact with the other groups which have a perception of these narratives so there's ingroup and outgroup Dynamics so it's a uh so it's a little bit again to how cells within an organism have their own boundaries but are also part of a larger um organism um so yeah I think that that does that answer the question from the person who asked that question think it's nice so perhaps a and Colin feel free to give in your marks and then we'll pass around the instructors it can be as long or as short as you like yeah okay may I go yeah uh first so uh I would like to say that uh the the key issue we are like uh trying to grasp is that effective inference uh has social and political implications due to its scale-free structure and how we can apply to to the social world then there is the self referential question that happens which is what is the the what is the the structure of active influence as a social construction itself and a political construction itself and what are our own um like our own what what is our own political power and Social Power in this social and political world and this is the like the fact like thing that is going on like because afterward what is my own social and political implication in this thing and the more you you get your own agency and the agency of active influence the more it becomes like a lot to process and uh the more you you grasp your own agency and um and the power you have and the how to say that responsibility yeah you you have so yeah I guess this is one of the key points we're trying to to get here um yeah uh I guess that's the the point I I wanted to make and yeah to to finish with that um we the the the game we are trying to play is to try to find our own blind spots in this game like what is the limit of the model and does it have a self referential limit or not so yeah that's it yeah I would um I would endorse that and um that was yeah I feel the same way I I found that uh my reading of of in just in the last year or so of of active inference ideas I found it very empowering uh the ideas of of active inference in in relation to the the individual mind uh are quite well developed and um it's yeah very empowering to be able to look at yourself through the active inference Paradigm and understand the thought processes that is you uh it's a very empowering thing and uh as uh as I was I was saying just recently starting to look at the idea that we can incorporate an active inference concept of the way the social world around us is structured which is part of that Paradigm too I think is uh is potentially more empowering but uh it's early days yet uh but I look forward very much to uh a more developed world where um people have discussions with each other not just about what I'm thinking and how I think but how I think about society and how we think about Society together and it's

embedded within that uh with that within that Paradigm I feel that's potentially something which is very empowering and I and I look forward to it and and I thank the speakers all for their contributions to this uh this series of talks thank you Colin so with the exception of Ben whose lecture and contribution kicked off this course and were excellent also we have the others of the instructors here so how do each of you want to uh kind of conclude chapter yeah Ma Ben L Rael yes I think uh of entities which have the capability of interacting to maximize the information they gather from the environment such that as a whole they can self evidence to um maintain the phenotypes they believe to be similar to their own and therefore their themselves um and how this leads to over time the sedimentation of norms and activities and eventually the reference to these norms and activities through the semantics and the scripts um and effectively how this leads us to all the complexities of language and how we may now be at the point where we have a simulacra that represents nothing it's just we just have these um these shells of representations and these shells of semantics uh which potentially have lost their anchoring in reality so I think that was the through line of the course I hope we made a good case for it and how active inference allows us to model this entire path not only across scale but across time as well um so yeah that's how I think uh I'm going to close yeah I think this is me now yeah I really like the story as well that we are building throughout the whole uh course starting with a very general idea of what is active inference with Ben and then trying to like build up this Collective notion how we how we contain information in a certain way that it's always synchronized so this is always sort of meaningful that can be described in certain ways by the mathematics of active inference and then understanding these questions hard questions that we asking now like how we can bring new variables how can bring novelty into the system which something that we don't have an answer yet but I find that's really cool that we don't like hold back in asking that because that I think this that's the next step for AC like understanding this how complexification comes about how can we describe that process and yeah and thanks sael and Daniel for putting this together was pretty cool and I hope you guys also have enjoyed thank you Lorena AEL so uh thank you L you're welcome uh I think uh the put his finger on uh something quite deep as uh as usual which is that if we want to have a coherent theory of nature that is not based on just here are the things that can be and we we'll draw that that that um is compatible with OP Evolution we need to address self falsity we need to address like um not compal but um recursion and the possibility for recursive individuation of physical beings and that's tough and that's very tough because the the the basis of presentation is just at list uh array of

things I want to represent and I represent that but if things can just happen that are new and that are by definition not in the field of possibilities that preexisted then we we we we are there is no way we can represent that and uh there is there is um a fam of approach in Quantum information theory that tries to tackle that how to draw physics without um um being rolling in a specific St space and just I'd like to to to to expl explain the history of that so the math it draws from algebraic topology so we study surfaces but not by studying surfaces by setting transformation within those surfaces but then we remove the surfaces and we keep only the transformation over what I don't know but somehow we can I guess reconstruct uh coherent mathematical entity from that and I'd like us to take a look at how this look what the math look like so can you can we see the screen okay so this is the structure of the quantum information formulation of the FP uh are dimension of interaction between system a and system B uh what defines the scope and the mode of this interaction I don't know and this is a con coone diagram that represent an agent together with the representation that he develops and I could tell you the words that correspond to the thing within this diagram but I do not know what they mean so it would be a hard thing to do and the math that is capable to tackle the self fality like maybe because I I did not see any proof that it actually does in fact it's it's not tractable for humans and I exclude mathematical physics for from humans there because of the abstractness of what is going on and because of the lack of any abuse reference that from anything we in understand in this and the work that we have to do if we want this to move forward and to have a self-coherent theory of social evolution is to somehow entangle from that um concrete meaning things that apply to the world and conversely use the way that we can think of it as concrete inter ction in the world to further specify the mathematics and prove that it is actually self-coherent and applies to stuff and this is a big big work uh in fact and this is kind of exciting I guess what I wanted to say is uh active inference is dead Long Live active inference great well it was a fun course and this closing section and slide kind of brought this Unity between Chris fields's Quantum work and this social science which in the disciplinary Paradigm are seen as distant or with different systems of interest and you can't get there from here but in a scale-free first principles Paradigm it's not even just that we can hold them side by side it's that they are more than Juck deposited so it's been awesome and its own great experience to work with all of the instructors who shared their amazing and Youthful and energetic perspectives and we explored a new format in a new way we're going to process the proceedings of the lectures and the discussions plus any final submitted question and answers through the syllabus website bring that together into a Open Access publication of some kind

so for people who are interested in that kind of editorial or curatorial work the active inference Journal is for them and then in 2024 and Beyond let's see where active inference is see where social Sciences are and hopefully as people are interested in take it from there we threw a many fun ideas but I guess we'll leave it till then to find out all right thank you everybody bye